

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. Student counseling in higher education draws on heterogeneous institutional knowledge, including relational curricula, cohort-specific tuition schedules, statutory exemption criteria, scholarship policies, and narrative administrative regulations. This setting challenges uniform retrieval and reasoning pipelines because different knowledge forms may benefit from different representations and processing strategies. We present CTU-Chat, a centralized supervisor-routed, domain-specialized multi-agent system for Vietnamese university counseling. The supervisor directs each query to one of four specialists, which receive scoped knowledge sources, retrieval paths, and tools. Relational academic information is handled through a knowledge graph; structured tuition and scholarship information is supported by dedicated lookup and calculation functions; and narrative regulations are retrieved through hybrid document search. Evaluation shows 97.0% raw structured-supervisor routing accuracy and bounded execution. A 1,960-decision semantic-neighbor stress test compares full-registry and Top- k single-agent baselines with routed generic and specialist gates as the registry expands from 11 to 51 tools. The routed specialist is the only architecture whose end-to-end pass rate improves over this range (85.3% $\rightarrow$ 88.0%, $\Delta = +2.7$ pp), while every single-agent variant degrades. At 51 tools, the specialist exceeds the matched global Top-10 single agent in end-to-end completion by 9.3 percentage points (95% CI [+2.0, +17.3]) and uses 52.4% fewer input tokens than the full-registry monolithic baseline. The high-minus-low scaling interaction is positive for both selection (+6.0 pp) and end-to-end completion (+7.3 pp), though the confidence intervals approach zero. On a redesigned 100-query held-out set, the full retrieval stack achieves 0.7400 Hit@1, 0.8608 Recall@5, and 0.8333 MRR@10, while controlled generation reaches 0.6534 Context Recall, 0.8636 Faithfulness, and 0.8325 Source Recall. These findings suggest that domain-specific tool disambiguation and heterogeneous evidence allocation can support institutional counseling under the evaluated conditions, although their computational and scaling trade-offs remain configuration dependent.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) is a key higher education institution in the Mekong Delta, with nearly 49,000 undergraduate students in the 2025–2026 academic year [1]. At an institution of this scale, advisory needs span a wide range of academic and administrative areas. Relevant regulations and policies may also change over time. Prior research on automated advisory systems includes CAAS, which supports chat- and voice-based admission counseling for information-technology programs [10], and REBot, which focuses on academic-regulation advising through dense retrieval and category-guided graph reasoning [9]. Their reported evaluations illustrate the value of task-specific retrieval and processing strategies, but remain centered on individual advisory domains. This motivates CTU-Chat as a coordinated multi-domain architecture for heterogeneous university counseling.

University counseling presents challenges not only because it covers multiple domains, but also because institutional knowledge is represented in different forms. (1) **Academic curricula** contain hierarchical structures and prerequisite relationships; (2) **tuition schedules** are organized by admission cohort and field of study; (3) **tuition exemption policies** and (4) **scholarship policies** combine eligibility conditions with specific support rates and thresholds; while (5) **general administrative regulations** are presented in lengthy narrative documents. Therefore, a single representation or retrieval method may not be equally suitable across all domains, motivating representation-specific evidence acquisition in CTU-Chat.

RAG and GraphRAG augment LLMs with external documents and knowledge graphs [8,5,3], while agentic architectures support tool invocation and multi-agent coordination [20,18,16]. However, these approaches do not by themselves determine how heterogeneous institutional evidence should be represented, routed, and processed. Moreover, known limitations in language-model arithmetic [14] motivate deterministic calculation for rule-based financial queries.

These challenges highlight three open research gaps in automated university counseling: (1) limited domain-specialized multi-agent orchestration; (2) uniform representation of heterogeneous institutional knowledge; and (3) the lack of representation-aware evidence routing. These gaps are examined in greater detail in Section 2.

The objective of this study is to develop and evaluate CTU-Chat for heterogeneous university counseling. To address the identified gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Tool Disambiguation and Scalability):**
How do full-registry and Top- k single-agent baselines compare with supervisor-routed generic and specialist gates in tool-selection reliability, end-to-end

completion, and computational cost as semantically overlapping within-domain tools expand the registry from 11 to 51 tools?

- **RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations):** What differences in context and answer quality are observed between the full heterogeneous configuration, uniform retrieval baselines, and its component ablations?
- **RQ3 (Representation-Aware Retrieval Effectiveness):** How does the complete representation-aware evidence configuration compare with sparse, dense, hybrid, and reranked baselines in retrieval effectiveness, both overall and across counseling domains?

We hypothesize that domain-scoped candidate retrieval combined with contrastive specialist policies can reduce confusion among semantically adjacent tools relative to a matched generic single-agent gate. This hypothesis concerns tool disambiguation under the evaluated registry expansions; it does not assume that multi-agent execution will dominate a full-registry monolithic baseline on every accuracy or efficiency measure. To investigate this hypothesis and the research questions above, we propose CTU-Chat, which coordinates four domain-specialized agents for academic, financial, scholarship, and general administrative inquiries within a unified framework combining graph-based retrieval, structured data access, deterministic calculation, and hybrid document retrieval.

Specifically, this work provides three primary contributions:

- **C1 (Domain-Specialized Multi-Agent Architecture):** We design and implement a centralized supervisor-routed architecture coordinating four bounded specialists with isolated evidence representations, deterministic route repair, and partitioned tool privileges, and evaluate its tool-disambiguation behavior in registries of up to 51 tools.
- **C2 (Heterogeneous Knowledge Allocation):** We develop a hybrid representation strategy that models relational curricula via property graphs, binds statutory fee and exemption lookup to deterministic arithmetic tools, and indexes narrative regulations through parent-resolved document retrieval.
- **C3 (Empirical Scalability and End-to-End Evaluation):** We evaluate semantic-neighbor tool ambiguity across 1,960 decisions with matched baselines, retrieval component stacking across 5 configurations, and generation quality across 7 configurations (2, 100 turns) on a redesigned 100-query held-out benchmark.

The remainder of this paper is organized as follows. **Section 2** reviews related work and summarizes the research gaps. **Section 3** presents the CTU-Chat architecture, evidence-acquisition methods, and domain-specialized agents. **Section 4** reports the experimental setup, results, and discussion. Finally, **Section 5** concludes the paper and outlines future work.

2 Related Work

2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11]; 85% of Vietnamese word types contain at least two syllables. While domain-adapted bi-encoders like PhoBERT capture semantic paraphrasing, Vietnamese higher education decrees rely heavily on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to mitigate both semantic drift and alphanumeric recall failures.

2.2 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [15] scores documents via term frequency (tf) and inverse document frequency (IDF) over an inverted index ($k_1 = 1.5, b = 0.75$). BM25 provides reliable exact token matching for rigid alphanumeric identifiers (decision codes, cohort labels, course codes), but cannot capture semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious or hallucinated relationships, especially when narrative documents lack explicit relational structure. Therefore, this work adopts selective rather than universal graph construction.

2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.6 Educational and University Counseling Systems

Educational chatbots assist in advising and learning support [17], with recent systems like REBot [9] combining text retrieval with category-guided GraphRAG for academic regulations. CTU-CHAT departs from prior advisory systems across three key aspects: (1) *Task Scope*: while representative prior advisory systems often emphasize single-domain text-QA or focus primarily on narrative regulations, CTU-CHAT models 113 curricula as relational property graphs and incorporates deterministic arithmetic engines (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for cohort-dependent tuition, exemptions, and scholarship thresholds; (2) *Multi-Agent Routing*: rather than a monolithic classifier-guided pipeline, a LangGraph supervisor dynamically routes queries to four bounded specialists with isolated evidence spaces and asymmetric tool privileges; and (3) *Metric Incommensurability*: prior benchmarks evaluate traditional classification accuracy and binary Q&A F_1 -scores, precluding direct numerical comparison with CTU-CHAT, which evaluates end-to-end generative fidelity via Ragas and bounded multi-agent tool execution benchmarks.

2.7 Comparative Synthesis and Research Gaps

Synthesizing established retrieval and reasoning paradigms highlights operational trade-offs for university counseling [2,3,6,7,15,16,18,20]. While individual techniques address specific needs—lexical matching for exact regulatory tokens, dense embeddings for semantic paraphrasing, and GraphRAG for relational prerequisites—the representative approaches reviewed here do not jointly address the three requirements of multi-domain institutional advising.

This analysis identifies three operational gaps in automated university counseling: (1) *Limited evidence on bounded specialization under tool ambiguity*: monolithic pipelines and unconstrained agent meshes may lack explicit execution guardrails, while it remains unclear when domain-specific policies improve upon matched single-agent selection as semantically adjacent tools accumulate; (2) *Limitations of uniform representation*: placing relational curricula, tabular tuition matrices, and narrative legal text in a single retrieval space may fragment prerequisite evidence or introduce semantic drift; and (3) *Absence of representation-aware*

routing: the representative systems reviewed here do not directly address dynamic steering across isolated representation pathways while offloading statutory arithmetic to deterministic engines. These requirements motivate the centralized supervisor-routed architecture and matched tool-disambiguation evaluation detailed in Sections 3–4.

3 Proposed Model

This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a contract-checked supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Each *specialist* is configured with domain-specific prompts, a scoped evidence path, and a restricted tool registry.

3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

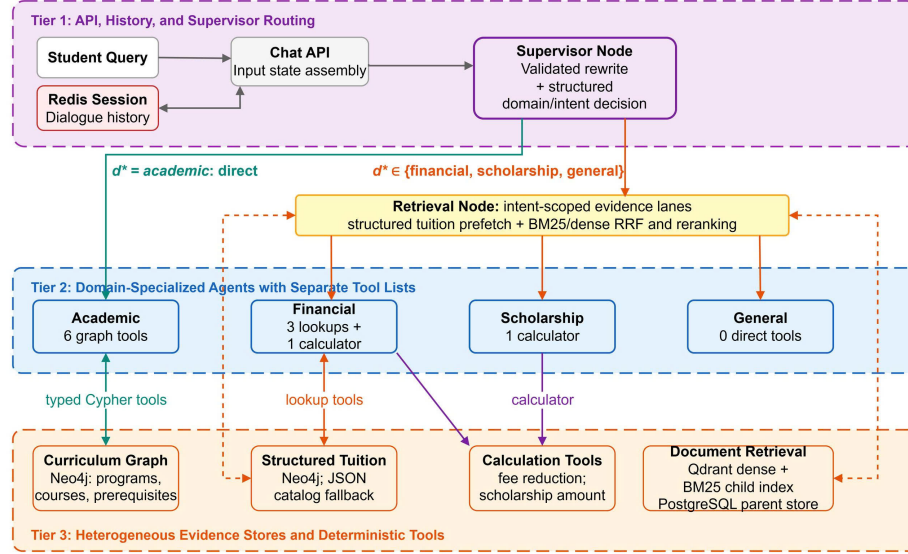


Fig. 1. Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked and may be repaired before conditional dispatch (repair is internal, not a separate node). Academic queries enter the graph-enabled agent, whereas other domains pass through intent-scoped retrieval. Solid and dashed links denote direct tool access and retrieval-node evidence acquisition, respectively.

Application-Level Specialization of LangGraph StateGraph: CTU-CHAT specializes LangGraph through typed state, an intent-owner contract, isolated paths, and partitioned tool registries. The contract checks structured supervisor output before dispatch; specialists then operate without peer delegation and see only their assigned tools.

- **Node-scoped state updates:** `AgentState` retains the validated rewrite, raw and repaired routes, repair reason, retrieved evidence, and response. The supervisor writes routing fields, retrieval writes `context` and `retrieval_instruction`, and the specialist writes `response`, making route repair traceable.
- **Two-Tier Conditional Routing Edges:** Instead of letting agents dynamically negotiate execution flow, the orchestration topology is governed by deterministic conditional functions:

$$\text{Edge}_{\text{sup}}(\mathcal{S}) = \begin{cases} \text{'academic_agent'}, & \text{if } \mathcal{S}[\text{'next_agent'}] = \text{'academic'} \\ \text{'retrieval'}, & \text{otherwise,} \end{cases} \quad (1)$$

bypassing document retrieval for queries assigned to the academic agent. At the retrieval exit, $\text{Edge}_{\text{ret}}(\mathcal{S})$ dispatches the retrieved context to the selected financial, scholarship, or general agent.

- **Acyclic outer orchestration:** The StateGraph has no peer-to-peer agent edges. Each selected domain-specialized agent returns to END, so the outer routing path cannot cycle. ReAct tool interactions inside an agent remain distinct from this outer graph topology.
- **Domain-specialized agents with asymmetric tools:** Academic, financial, and scholarship agents receive scoped tool lists; the general agent has no direct tool. A prompt gate requests one call only when required parameters are valid. If a tool-backed query omits an invocation, one bounded retry names the required tool; missing parameters instead trigger clarification.

3.2 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

Schema defined by the ingestion layer: The Neo4j graph models relational dependencies across academic counseling using a simplified schema, comprising over 1,500 entity nodes and 3,800 relational edges across 113 undergraduate curricula and statutory fee schedules. It incorporates 10 node labels (7 for curriculum: Program, Course, Block, etc.; and 3 for tuition: TuitionFee, TuitionPolicy, ExemptionBasisRate) interconnected via 13 relationship types (HAS_BLOCK, CONTAINS, REQUIRES, PARALLEL_WITH, HAS_PLO, HAS_PEO, REALIZED_BY, HAS_TRACK, BELONGS_TO_TRACK, HAS_JOB_POSITION, HAS_TUITION, GOVERNED_BY, and HAS_EXEMPTION_BASIS). Narrative regulations remain in the document store unless linked to structured entities.

Cypher tool mechanics: The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search:* `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal:* `xem_chuoi_tien_quyet` follows REQUIRES paths up to ten hops and returns direct PARALLEL_WITH courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup:* parameterized queries match TuitionFee, TuitionPolicy, and ExemptionBasisRate properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. The multi-agent workflow therefore incorporates graph evidence through bounded operations, reducing exposure to the syntactic and semantic errors associated with unconstrained Cypher generation.

3.3 Algorithmic Foundations: Routing and Deterministic Arithmetic

Algorithm 1 specifies the supervisor-routing and evidence-path dispatch procedure.

Algorithm 1: Supervisor Routing and Conditional Evidence-Path Dispatch

Input: Student query q , Dialogue history $\mathcal{H}$, Shared state $\mathcal{S}$, Ownership contract $\mathcal{P}$

Output: State updates and next graph node

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2  $q_{\text{prev}} \leftarrow \text{MostRecentUserQuery}(\mathcal{H})$ ;
3 if ShouldRewriteQuery( $q, q_{\text{prev}}$ ) then
4    $q_{\text{cand}} \leftarrow \text{FastLLM.Rewrite}(q, q_{\text{prev}})$ ;
5   if ValidateRewrite( $q, q_{\text{cand}}, q_{\text{prev}}$ ) then
6      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
7  $\langle d_{\text{raw}}, \iota_{\text{raw}} \rangle \leftarrow \text{StructuredSupervisor}(q)$ ;
8 if structured classification fails then
9    $d_{\text{raw}} \leftarrow \text{'general'}$ ;  $\iota_{\text{raw}} \leftarrow \text{'other'}$ ;
10  $\langle d^*, \iota^*, \rho \rangle \leftarrow \text{RepairRoute}(q, d_{\text{raw}}, \iota_{\text{raw}}, \mathcal{P})$ ;
11 Write  $q_{\text{standalone}}, d_{\text{raw}}, \iota_{\text{raw}}, d^*, \iota^*, \rho$  to  $\mathcal{S}$ ;
12 if  $d^* = \text{'academic'}$  then
13    $v_{\text{next}} \leftarrow \text{'academic\_agent'}$ ;
14 else
15   if  $d^* = \text{'financial'}$  then
16     Prefetch tuition evidence from Neo4j, else JSON;
17   Build retrieval lanes from  $\iota^*$  and retrieve evidence;
18    $v_{\text{next}} \leftarrow d^*_{\text{-agent}}$ ;
19 return  $\langle \mathcal{S}, v_{\text{next}} \rangle$ ;

```

Rewrites require validation, and structured classification falls back to *general/other*. **RepairRoute** preserves valid predictions but corrects strong rule conflicts or incompatible agent–intent pairs under $\mathcal{P}$. Conditional edges send academic queries to the graph agent and other domains through retrieval.

Deterministic Tuition Reduction Arithmetic: For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic tool `tinh.toan.hoc.phi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition T , the per-credit exemption basis B , and the percentage entitlement p —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (2)$$

where D is the reduction per credit and P is the remaining amount per credit. Monetary inputs are expected in VND per credit. The tool enforces $0 \leq p \leq 100$ and clamps P at zero; values outside the percentage range return an error.

Rather than a new general-purpose algorithm, the contribution lies in composing validated rewriting, constrained routing, representation-specific dispatch, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT preserves relational curriculum knowledge in Neo4j, accesses cohort-dependent tuition data through structured lookup, and retains narrative regulations and scholarship policies in document retrieval.

Table 1 makes this policy explicit by separating the physical evidence representation from the mechanism used to access it at runtime.

Table 1. Decision criteria mapping query characteristics to evidence representation mechanisms.

Query characteristic	Evidence path	Rationale
Program/course attributes	Neo4j property lookup	Exact or normalized match
Prerequisites/shared courses	Neo4j traversal	Stored graph relationships
Regulation identifiers	BM25 child index	Exact token match
Paraphrased regulations	Hybrid RRF + reranker	Lexical + semantic signals
Fee/scholarship calculation	Python tool	Fixed formula outside LLM

At runtime, academic queries enter the graph agent directly; other domains enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback. The repaired intent then selects metadata-filtered document lanes, and packed evidence is passed only to the selected specialist.

Hybrid Document Retrieval Path. For narrative regulations and scholarship criteria, evidence is acquired through the following pipeline:

1. *Child-level candidate generation.* A BM25 index searches child chunks, while Qdrant performs dense retrieval over their 768-dimensional embeddings.
2. *Parent resolution.* Parent identifiers associated with the retrieved child chunks are used to load the corresponding parent sections from PostgreSQL.
3. *Reciprocal Rank Fusion.* The dense- and BM25-derived parent rankings are combined as

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60.$$

4. *Cross-encoder reranking.* The fused parent candidates are reranked using BAAI/bge-reranker-v2-m3. In the evaluated full configuration, candidates from intent-governed lanes are pooled with the top 20 candidates from

unfiltered hybrid retrieval and the top-ranked BM25 candidates before deduplication. Candidates with reranker scores below 0.20 are removed unless fewer than two remain. Graph evidence is then added under predefined quotas, and up to seven passages are packed for generation.

Parent sections use a 2,800-character target with 100-character overlap; child chunks use a 400-character target with 50-character overlap.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT.

4.1 Experimental Benchmark and Protocols

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded semi-automated protocol over official Can Tho University (CTU) statutes. Candidate queries with diverse conversational forms were generated from statutory articles and reviewed for ground-truth intents, passages, required facts, and answers. The 100-query development split supports index calibration, prompt engineering, and failure analysis. The separately constructed 100-query held-out set follows a predefined complexity distribution of 40 direct single-hop, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries. It includes 51 formal and 49 colloquial formulations; 56 queries have one gold source and 44 require multiple gold sources.

An audit using entity inventories separate from the development split found no exact cross-split duplicates and a maximum 5-gram Jaccard similarity of 0.1875. Gold sources and required facts were reviewed against 244 institutional documents and 559 structured tuition records. The held-out set contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains and are analyzed separately. Tables 2 and 4 report results on this held-out set.

The final run (ID 20260921T043243Z) used Ragas 0.4.3 with Answer Relevance strictness 1 and 10,000 bootstrap resamples for 95% confidence intervals. Its coverage audit found all 10,500 required metric values valid, with no missing values or evaluation failures.

Evaluated Model and Infrastructure: The evaluated run used the Gemini 2.5 Flash Lite model identifier through Google Cloud Vertex AI with temperature 0.0; later deployment-model substitutions were not evaluated. Each query was evaluated in three repetitions, producing 2,100 generated answers and 2,100 corresponding Ragas evaluations (7 configurations $\times$ 100 queries $\times$ 3 repetitions). Neural reranking used an NVIDIA GeForce GTX 1650 GPU (4 GB VRAM). Neo4j 5.20 with APOC stored relational curricula; PostgreSQL stored parent chunks (2,800 characters, 100 overlap); and Qdrant indexed child chunks (400 characters, 50 overlap) using a 768-dimensional `vietnamese-bi-encoder`. The system used `bge-reranker-v2-m3`, LangGraph StateGraph, and Redis 7.

Scenario 3 Semantic-Neighbor Protocol: The tool-ambiguity experiment fixes 50 production cases (30 academic and 20 financial) while expanding an 11-tool core with 40 evaluation-only, operationally plausible semantic neighbors distributed across ten production tool families. Adding zero, two, or four neighbors per family produces registries of 11, 31, and 51 tools. Four configurations are compared: a monolithic full-registry gate, a global Top-7 single-agent gate, a supervisor-routed Top-7 gate with a generic selection policy, and a supervisor-routed Top-7 specialist gate with contrastive use/avoid rules derived from the tool contracts. Each primary case is evaluated under three counterbalanced tool-order blocks, yielding 1,800 decisions. A separate 40-case coverage suite makes every semantic-neighbor tool the gold target once at the 51-tool level, adding 160 decisions and producing 1,960 total records. To isolate downstream selection, 190 live supervisor decisions are shared across routed configurations and ambiguity levels for the same case and order block. Paired differences and high-minus-low interactions use 10,000 case-level bootstrap resamples after averaging repetitions within each case.

4.2 Evaluation Metrics

Evaluation Metrics: Scenario 1 reports source-document-level Hit@ k ($k \in \{1, 3\}$), Precision@5, Recall@5, MRR@10, and domain-level Hit@1. Scenario 3 measures routing accuracy, Macro-F1, intent accuracy, tool selection and argument Exact Match (EM), end-to-end pass rate, adversarial tool gating, and safe behavior under `max_tool.calls = 1` and a 60-s timeout. The semantic-neighbor analysis additionally reports gold-tool shortlist recall, visible tools, input tokens, decision latency, paired 95% confidence intervals, and the high-minus-low interaction in paired differences.

Scenario 2 evaluates Top-7 synthesis using Ragas [4]: Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), Answer Correctness (AC), and Faithfulness (Faith). Factual Exact Match (Fact EM) measures strict numerical tuition-fee and credit agreement, while Source Recall measures gold-source coverage and Source Average Precision (Source AP) evaluates the ranking of retrieved gold sources. To reduce LLM-judge bias [21,4], the evaluation uses gold reference answers and source documents for reference-grounded metrics, temperature-zero decoding over three repetitions, and programmatic Fact EM for tuition arithmetic.

4.3 Experimental Scenarios and Benchmark Results

We evaluate CTU-CHAT through retrieval component stacking (RQ3), end-to-end generation and ablation (RQ2), and multi-agent coordination and tool reliability (RQ1).

Scenario 1 — Retrieval Component Stacking (E1–E5): Scenario 1 compares five cumulative retrieval configurations, from sparse and dense baselines to

hybrid fusion, neural reranking, and the proposed intent-guided evidence stack, at aggregate and complexity-stratified levels on the 100-query held-out set.

Table 2 shows that E4 and E5 exceed the sparse, dense, and hybrid baselines on every aggregate metric. E4 reaches 0.7000 Hit@1 and 0.8054 MRR@10. The full intent-governed configuration E5 records the highest values: 0.7400 Hit@1, 0.9200 Hit@3, 0.3020 Precision@5, 0.8608 Recall@5, and 0.8333 MRR@10. Its higher latency (21.86 s vs. 15.37 s for E4) is consistent with additional intent-governed retrieval and graph-processing stages.

Table 2. Scenario 1: Progressive retrieval component stacking on the redesigned Held-Out Test Set ($N = 100$).

Retrieval Configuration	Hit@1	Hit@3	P@5	Recall@5	MRR@10	Latency (ms)
E1: BM25 Lexical Baseline	0.5300	0.7700	0.2060	0.6167	0.6695	7.08
E2: Dense Vector Semantic	0.4600	0.6500	0.1760	0.5242	0.5604	33.66
E3: Hybrid RRF (BM25 + Dense)	0.5200	0.7900	0.2240	0.6625	0.6725	37.57
E4: Hybrid + Neural Reranker	0.7000	0.9100	0.2540	0.7508	0.8054	15,366.33
E5: CTU-Chat Full Retrieval	0.7400	0.9200	0.3020	0.8608	0.8333	21,857.56

Table 3. E4–E5 retrieval results by held-out query complexity.

Complexity Tier	N	Hit@1		Recall@5	
		E4	E5	E4	E5
Direct single-hop	40	0.6500	0.6750	0.8500	0.9250
Within-domain multi-hop	20	0.7000	0.7000	0.7000	0.7583
Cross-domain	20	0.7500	0.8000	0.6917	0.8250
Entity comparison	10	0.8000	0.9000	0.6667	0.9500
Temporal	5	0.8000	1.0000	0.9000	1.0000
Adversarial	5	0.6000	0.6000	0.4167	0.5833

E5 matches or exceeds E4 on both metrics in every complexity tier. The largest Recall@5 differences occur for entity-comparison (28.33 percentage points; 0.9500 vs. 0.6667) and cross-domain queries (13.33 percentage points; 0.8250 vs. 0.6917). For the five temporal queries, E5 obtains 1.0000 on both metrics.

Scenario 2 — Controlled Retrieval-Conditioned Generation and Modular Ablation (T1–T7): Scenario 2 compares the full governed multi-evidence configuration (T4) with three retrieval baselines (T1–T3) and three ablations (T5–T7) over 2,100 evaluation turns. As shown in Table 4, T4 exceeds BM25 on all eight metrics, including higher Faithfulness (0.8636 vs. 0.8042) and a 4.07-percentage-point Fact EM difference (63.31% vs. 59.24%). T4 records the highest Context Recall (0.6534), Faithfulness (0.8636), Source Recall (0.8325), and Source AP (0.7316). T6 leads Context Precision, Answer Relevancy, and Answer Correctness, while T7 obtains slightly higher Fact EM (63.45%).

Table 4. Scenario 2: Controlled retrieval-conditioned generation and modular ablation on the redesigned Held-Out Test Set ($N = 100, R = 3$, 2,100 total evaluation turns, Top-7 context).

Cfg.	CP	CR	AR	AC	Faith	Fact EM	Src. R	Src. AP
T1	0.2522	0.5349	0.3901	0.4615	0.8042	0.5924	0.6358	0.4637
T2	0.1626	0.2648	0.2370	0.3372	0.5817	0.5179	0.5342	0.4094
T3	0.2071	0.4693	0.3972	0.4361	0.7486	0.5792	0.6892	0.5015
T4	0.3511	0.6534	0.4468	0.4785	0.8636	0.6331	0.8325	0.7316
T5	0.2248	0.5192	0.3425	0.4330	0.7618	0.6056	0.7892	0.6834
T6	0.3765	0.5791	0.4719	0.4913	0.8376	0.6184	0.6925	0.5939
T7	0.3541	0.6299	0.4304	0.4785	0.8427	0.6345	0.8208	0.7277

Configurations: T1: BM25; T2: dense; T3: hybrid RRF; T4: full governed multi-evidence configuration; T5: without cross-reranker; T6: without Neo4j grounding; T7: without subspace governance. *Metrics:* Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Faith: Faithfulness; Src. R: Source Recall; Src. AP: Source Average Precision. Ragas values are means over 300 query–repetition observations per configuration; corresponding standard deviations and bootstrap confidence intervals are retained in the experiment artifacts. All metrics are normalized to $[0, 1]$; higher is better.

Scenario 3 — Multi-Agent Routing, Specialist Tool Reliability, and Semantic Tool Ambiguity: Scenario 3 evaluates coordination, bounded safety, and selection among semantically adjacent tools (Table 5). Panel A scores raw supervisor output before route repair; Panels B–C test isolated specialist gates and adversarial inputs. Panel D reports the 1,800-decision fixed-case semantic-neighbor experiment across 11, 31, and 51 tools; the additional 160-decision coverage suite is reported in the accompanying analysis.

4.4 Discussion

Answering RQ1: Before production route repair, the evaluated raw structured supervisor achieves 97.0% routing accuracy, 97.8% Macro-F1, and 95.0% intent accuracy, compared with 85.0%, 69.2%, and 69.0% for rule-based routing. The isolated deterministic tuition tools achieve 100.0% end-to-end pass rates, whereas scholarship calculation reaches 76.7% with 86.7% tool selection; all tested paths remain bounded. Panel D evaluates scalability under increasing intra-domain semantic-neighbor density. The routed specialist is the only architecture whose end-to-end pass rate *improves* as the registry scales from 11 to 51 tools (85.3% $\rightarrow$ 88.0%, $\Delta = +2.7$ pp), whereas every single-agent variant degrades: full-registry single 86.0% $\rightarrow$ 83.3% ($\Delta = -2.7$ pp), global Top-10 83.3% $\rightarrow$ 78.7% ($\Delta = -4.6$ pp). At the densest registry (51 tools), the specialist exceeds the matched global Top-10 single agent in selection by +6.0 pp (95% CI $[-1.3, +14.0]$) and in end-to-end completion by +9.3 pp (CI $[+2.0, +17.3]$). The specialist also exceeds the routed generic gate in E2E by +5.3 pp (CI $[+0.7, +10.7]$) across all three registry sizes, consistent with a contribution from the contrastive domain policy.

Table 5. Scenario 3: Routing, specialist tool reliability, adversarial robustness, and semantic-neighbor tool ambiguity under a maximum of one tool call per decision ($\text{max_tool_calls} = 1$).

Panel A: Raw Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded	Routing Paradigm	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	Heuristic Regex	
Raw LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	Structured Pydantic	
Panel B: Specialist Tool Execution Reliability ($N = 90$ runs across 30 test cases, 3 repeated runs)							
Specialist Tool Function	Runs (N)	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Paradigm
tuition_lookup	30	100.0%	100.0%	100.0%	100.0%	100.0%	Structured Catalog
scholarship_calculation	30	86.7%	76.7%	86.7%	76.7%	100.0%	LLM Tool Selection
tuition_reduction_calculation	30	100.0%	100.0%	100.0%	100.0%	100.0%	Arithmetic Tool
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate Safe	Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	–
Panel D: Semantic-Neighbor Tool Ambiguity ($N = 50$ fixed cases, 3 reps, $N = 1,800$)							
Registry	Architecture	Visible	Gold@10	Selection	E2E Pass	Input Tok.	Latency (ms)
11 tools	Full-registry single	11.0	–	94.0%	86.0%	1,663	1,098
	Global Top-10 single	10.0	100.0%	93.3%	83.3%	1,541	1,148
	Routed generic	5.1	98.0%	90.7%	80.0%	1,204	1,889
	Routed specialist	5.1	98.0%	93.3%	85.3%	1,441	1,983
31 tools	Full-registry single	31.0	–	91.3%	84.7%	2,706	1,157
	Global Top-10 single	10.0	98.0%	90.0%	81.3%	1,149	1,033
	Routed generic	9.8	98.0%	88.0%	79.3%	1,396	1,958
	Routed specialist	9.8	98.0%	93.3%	88.7%	1,830	2,015
51 tools	Full-registry single	51.0	–	92.7%	83.3%	3,722	1,289
	Global Top-10 single	10.0	96.0%	86.7%	78.7%	1,036	997
	Routed generic	9.8	98.0%	91.3%	82.7%	1,340	1,907
	Routed specialist	9.8	98.0%	92.7%	88.0%	1,770	2,021

Note: Panels A–C evaluate raw routing, isolated tool gates, and adversarial robustness. Panel D uses 50 fixed production cases across 11, 31, and 51 tools with a distilled supervisor prompt. Gold@10 is gold-tool shortlist recall at $k = 10$; Visible is the mean active tool count. Paired intervals use 10,000 case-level bootstrap resamples. The separate coverage suite makes each of 40 semantic-neighbor tools the gold target once.

Selection accuracy remains near-constant for the specialist ($93.3\% \rightarrow 92.7\%$, $\Delta = -0.6$ pp) while the Top-10 single agent drops 6.6 pp over the same range.

The token-efficiency difference is noteworthy. The distilled supervisor reduces the routing overhead to approximately 500 tokens, bringing the specialist’s total input tokens to 1,770 at 51 tools—52.4% fewer than the full-registry single agent (3,722). As the registry grows $4.6\times$, the specialist’s token cost increases by a factor of $1.23\times$, compared with the monolithic agent’s $2.24\times$ ($O(N)$) growth. The high-minus-low interaction (specialist vs. Top-10) is $+6.0$ pp for selection (CI $[-0.0, +13.3]$) and $+7.3$ pp for E2E (CI $[-0.0, +15.3]$); both are positive and the confidence intervals approach but do not cross zero. In the coverage suite at 51 tools, the specialist matches the full-registry single in selection (92.5%) and exceeds it in E2E (82.5% vs. 80.0%), despite seeing only 9.8 of 51 tools per query. Conditional on the gold tool being shortlisted, specialist selection reaches 94.6%. Taken together, the evidence is consistent with a scalability advantage for domain-partitioned routing: under the evaluated conditions, contrastive specialist gates maintain or improve both accuracy and token efficiency as the tool registry grows, whereas monolithic and unstructured Top- k approaches degrade under increasing semantic-neighbor density.

Answering RQ2: T4 exceeds the no-cross-reranker configuration T5 on every reported metric. The paired Ragas differences favor T4 for all five judged metrics; for example, T4’s Faithfulness is 0.1018 higher than T5’s (95% bootstrap CI [+0.0548, +0.1475]). Comparisons with the other ablations are mixed. Relative to T6 (without graph evidence), T4 has higher Context Recall, Faithfulness, Source Recall, and Fact EM, but lower Context Precision, Answer Relevancy, Answer Correctness, and Source AP. Relative to T7 (without governance), T4 has higher Context Recall, Faithfulness, Answer Relevancy, Source Recall, and Source AP, while T7 is slightly higher on Context Precision and Fact EM; their paired Ragas confidence intervals all include zero. These results support component-specific associations rather than uniform causal effects.

Answering RQ3: E4, which adds neural reranking to hybrid retrieval, exceeds E3 on every aggregate metric. E5 records the highest aggregate values and matches or exceeds E4 on Hit@1 and Recall@5 in every complexity tier. Because E5 jointly adds intent-governed lanes, graph evidence, and quota-based packing, these differences cannot be attributed to an individual component; they are nevertheless consistent with improved multi-source evidence aggregation in the evaluated setting.

4.5 Threats to Validity and Limitations

The evaluation is limited to CTU and 100 held-out queries, with five temporal and five adversarial cases; Gemini 2.5 Flash Lite may also introduce generator–judge bias. Scenario 3 uses 50 fixed cases and evaluation-only semantic-neighbor contracts. Its coverage suite has one repetition, and shared supervisor decisions control downstream comparisons but omit independent route variability. While the high-minus-low scaling interaction is positive, its confidence intervals approach but do not exclude zero, so the differential degradation rate warrants replication with larger sample sizes. The routed specialist incurs approximately $2\times$ higher latency than the single-agent alternatives due to the two-hop LLM call sequence (supervisor then gate), despite achieving lower total token cost than the full-registry monolithic agent. One-decision gates do not reproduce the complete repair/retry path. Live fault injection, naturally evolving registries, and multi-institutional validation remain future work.

5 Conclusion and Future Work

This study examined centralized supervision and representation-specific evidence paths for heterogeneous university counseling. Raw supervisor output reaches 97.0% agent accuracy. In the semantic-neighbor study, the routed specialist is the only architecture whose end-to-end pass rate improves as the registry scales from 11 to 51 tools (85.3% $\rightarrow$ 88.0%, $\Delta = +2.7$ pp), while the full-registry single agent degrades by 2.7 pp and the Top-10 single agent by 4.6 pp over the same range. At 51 tools, the specialist exceeds the matched Top-10 single agent in end-to-end completion by 9.3 pp (95% CI [+2.0, +17.3]) and consumes 52.4% fewer input

tokens than the full-registry monolithic baseline (1,770 vs. 3,722). The high-minus-low scaling interaction is positive for both selection (+6.0 pp) and end-to-end completion (+7.3 pp), with confidence intervals approaching but not excluding zero; replication with larger samples is warranted. Comparisons with the routed generic gate are consistent with a contribution from contrastive domain policies. On the 100-query held-out set, E5 records 0.7400 Hit@1, 0.8608 Recall@5, and 0.8333 MRR@10, while T4 leads Context Recall (0.6534), Faithfulness (0.8636), Source Recall (0.8325), and Source AP (0.7316); individual ablations lead other metrics.

These findings suggest that domain-partitioned routing with contrastive specialist gates can maintain or improve both accuracy and token efficiency as the tool registry grows under the evaluated conditions, while monolithic and unstructured Top- k approaches tend to degrade under increasing semantic-neighbor density. Heterogeneous evidence may benefit from representation-specific paths. The routed specialist incurs approximately $2\times$ higher latency than single-agent alternatives, and the scaling interaction, while positive, has not been confirmed at larger sample sizes. Future work will use anonymized logs and human and cross-institutional evaluation to test learned candidate retrieval, compressed routing, and independent routes at larger scales.

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